Lab Programs

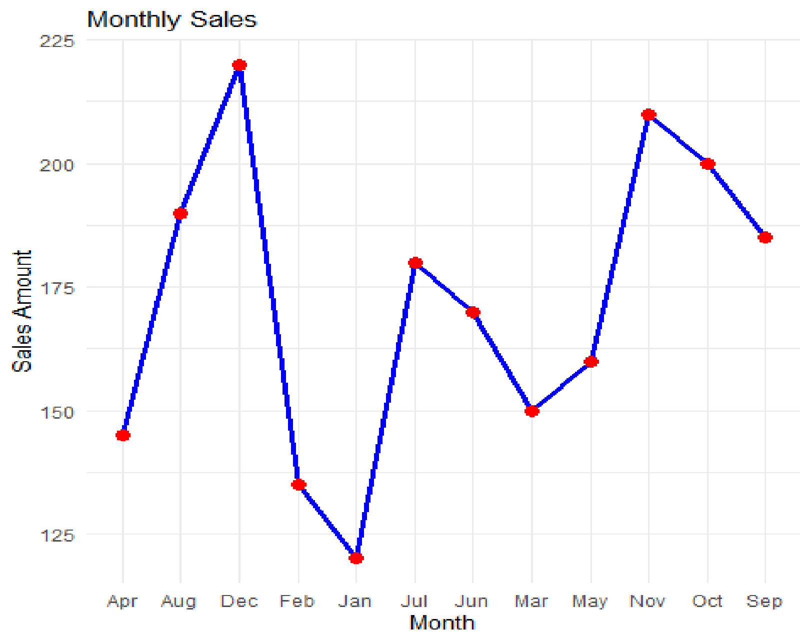
SET 1-Monthly Sales Data

1A. Line Chart

Program:

```
ggplot(sales_data,aes(x=Month,y=Sales,group=1))+
geom_line(color="blue",linewidth=1.2)+
geom_point(color="red",size=3)+
labs(title="Monthly Sales",x="Month",y="Sales Amount")+
theme_minimal()
```

Output:



1B. Bar Chart

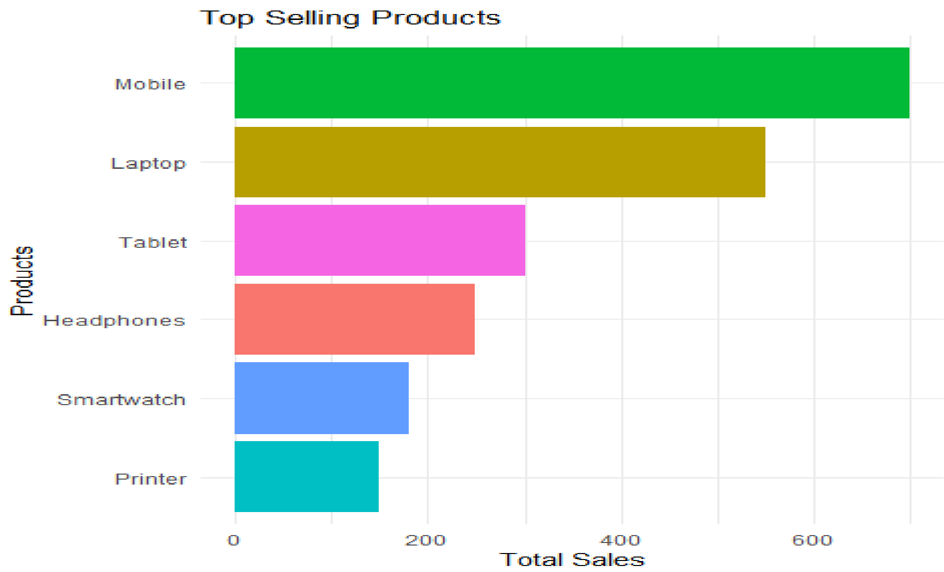
Program:

```
ggplot(product_data,aes(x=reorder(Product,Sales),y=Sales,fill=Product))+
geom_bar(stat="identity")+
coord_flip()+
labs(title="Top Selling Products",x="Products",y="Total Sales")+
```

theme_minimal()+

theme(legend.position="none")

Output:

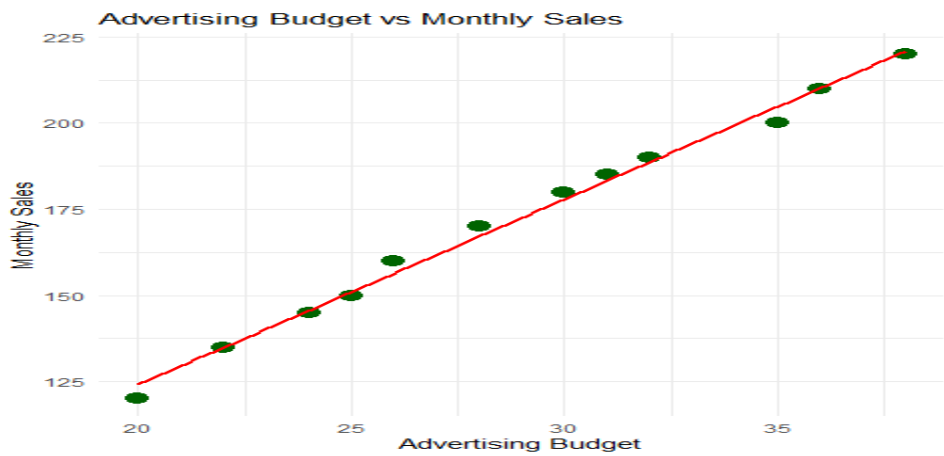


1C. Scatter Plot

Program:

```
ggplot(sales_data,aes(x=Advertising_Budget,y=Sales))+
geom_point(size=4,color="darkgreen")+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Advertising Budget vs Monthly Sales",
x="Advertising Budget",
y="Monthly Sales")+
theme_minimal()
```

Output:



1D. Combined Interactive Dashboard

Program:

```
line_plot <- ggplotly(
ggplot(sales_data,aes(x=Month,y=Sales,group=1))+
geom_line(color="blue")+
geom_point(color="red")
)
bar_plot <- ggplotly(
ggplot(product_data,aes(x=Product,y=Sales,fill=Product))+
geom_bar(stat="identity")+
theme(legend.position="none")
)
subplot(line_plot,bar_plot,nrows=2)
```

Output:

SET 2- Customer Feedback Analysis

2A. Histogram of Customer Ages

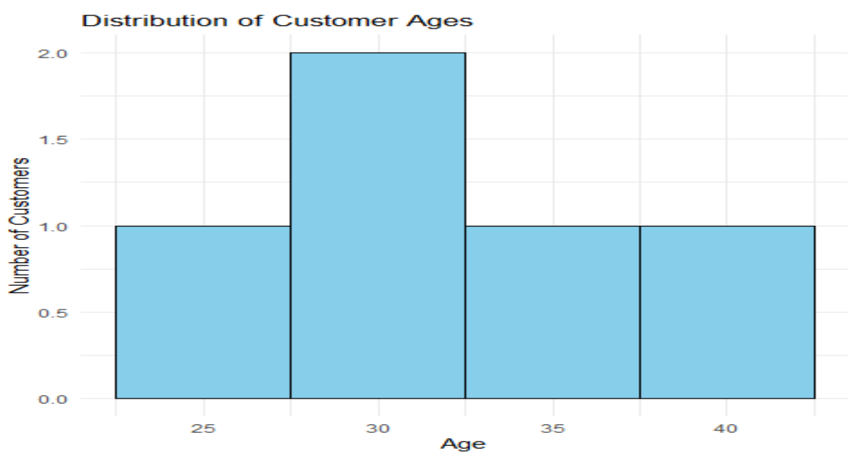
Step 1: Install Packages (Run Only Once)

```
install.packages("ggplot2")
install.packages("dplyr")
install.packages("wordcloud")
install.packages("RColorBrewer")
```

Program:

```
library(ggplot2)
customer_data <- data.frame(
CustomerID=c(1,2,3,4,5),
Age=c(25,30,35,28,40),
SatisfactionScore=c(4,5,3,4,5)
)
ggplot(customer_data, aes(x=Age)) +
geom_histogram(binwidth=5, fill="skyblue", color="black") +
labs(
title="Distribution of Customer Ages",
x="Age",
y="Number of Customers"
) +
theme_minimal()
```

Output:

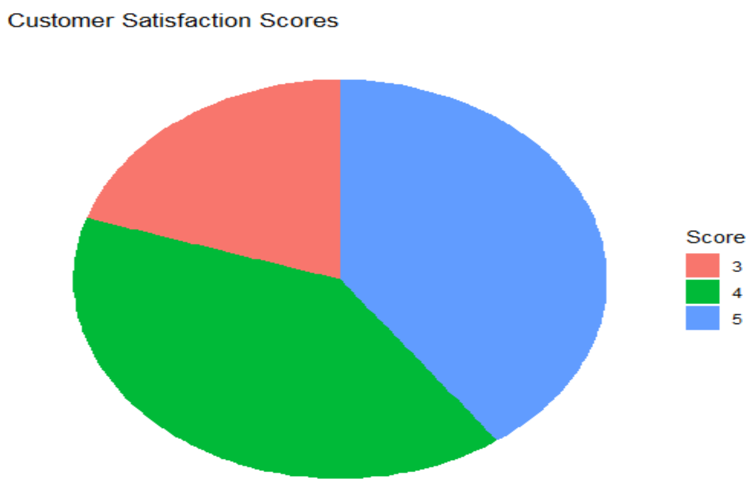


2B. Pie Chart of Satisfaction Scores

Program:

```
score_count <- customer_data %>%
count(SatisfactionScore)
ggplot(score_count,aes(x="",y=n,fill=factor(SatisfactionScore)))+
geom_bar(stat="identity",width=1)+
coord_polar("y")+
labs(title="Customer Satisfaction Scores",
fill="Score")+
theme_void()
```

Output:



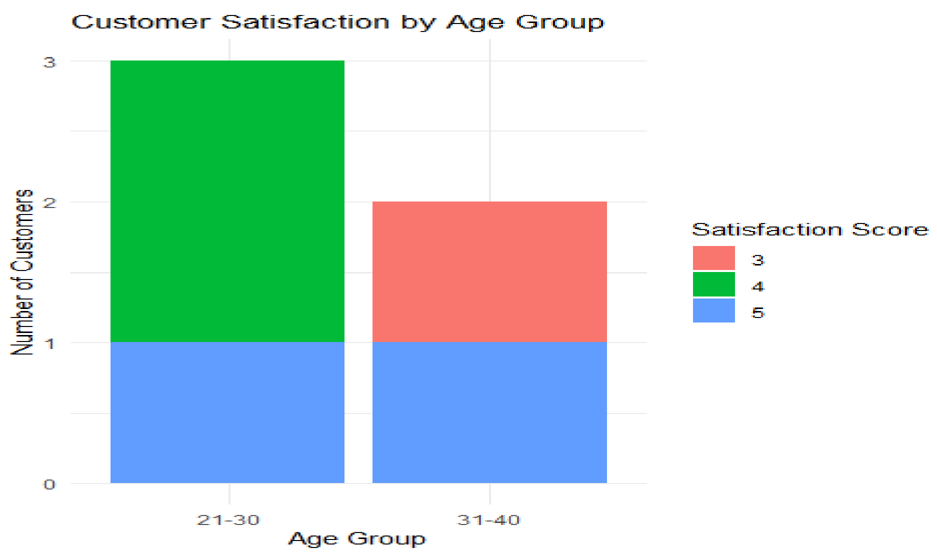
2C. Stacked Bar Chart by Age Group

Program:

```
customer_data$AgeGroup <- cut(customer_data$Age,
breaks=c(20,30,40,50),
labels=c("21-30","31-40","41-50"),
include.lowest=TRUE)
customer_data
ggplot(customer_data,
aes(x=AgeGroup,
fill=factor(SatisfactionScore)))+
```

```
geom_bar()+
labs(title="Customer Satisfaction by Age Group",
x="Age Group",
y="Number of Customers",
fill="Satisfaction Score")+
theme_minimal()
```

Output:



2D. Word Cloud

Program:

```
feedback <- c(
"Excellent service and friendly staff",
"Very satisfied with product quality",
"Good customer support and fast delivery",
"Excellent quality and quick response",
"Very happy with the service"
)
docs <- Corpus(VectorSource(feedback))
docs <- tm_map(docs, content_transformer(tolower))
docs <- tm_map(docs, removePunctuation)
docs <- tm_map(docs, removeNumbers)
docs <- tm_map(docs, removeWords, stopwords("english"))
```

```
docs <- tm_map(docs, stripWhitespace)
dtm <- DocumentTermMatrix(docs)
m <- as.matrix(dtm)
word_freq <- sort(colSums(m), decreasing=TRUE)
wordcloud(
names(word_freq),
word_freq,
min.freq=1,
colors=brewer.pal(8,"Dark2"),
random.order=FALSE,
scale=c(4,1)
)
```

Output:

Set 3- Employee Performance Evaluation.

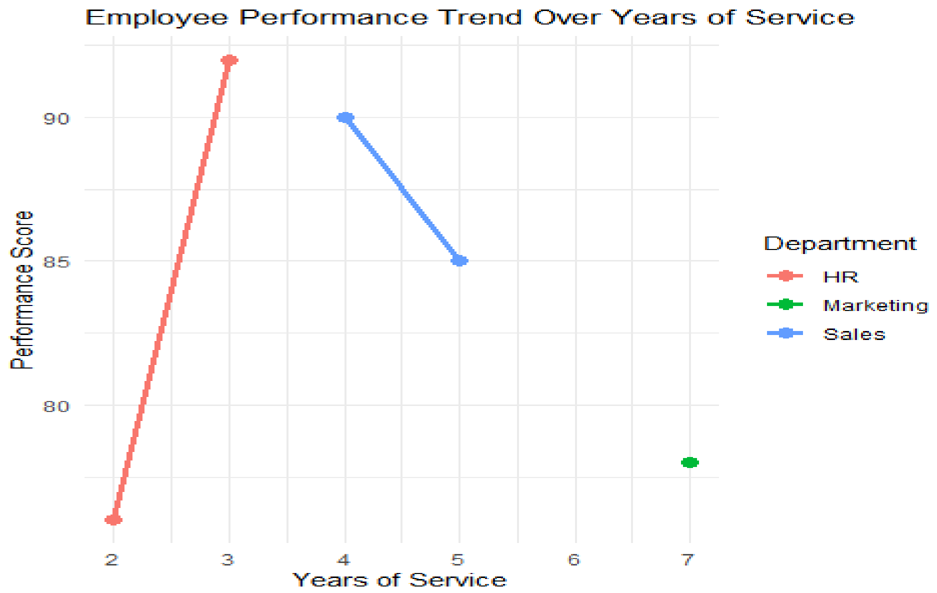
3A. Line Chart

Program:

```
employee_data <- data.frame(
EmployeeID=c(1,2,3,4,5),
```

```
Department=c("Sales","HR","Marketing","Sales","HR"),
YearsOfService=c(5,3,7,4,2),
PerformanceScore=c(85,92,78,90,76)
)
ggplot(employee_data,
aes(x=YearsOfService,
y=PerformanceScore,
color=Department,
group=Department))+
geom_line(linewidth=1.2)+
geom_point(size=3)+
labs(title="Employee Performance Trend Over Years of Service",
x="Years of Service",
y="Performance Score",
color="Department")+
theme_minimal()
```

Output:



3B. Bar Chart (Department Distribution)

Program:

```
ggplot(employee_data,
```

aes(x=Department,

fill=Department))+

geom_bar()+

labs(title="Employee Distribution by Department",

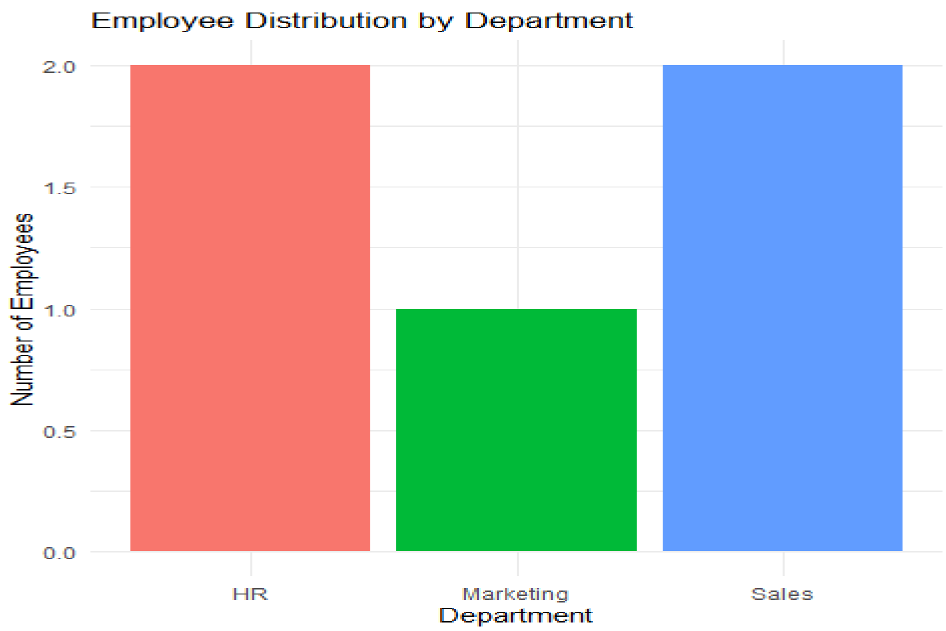
x="Department",

y="Number of Employees")+

theme_minimal()+

theme(legend.position="none")

Output:



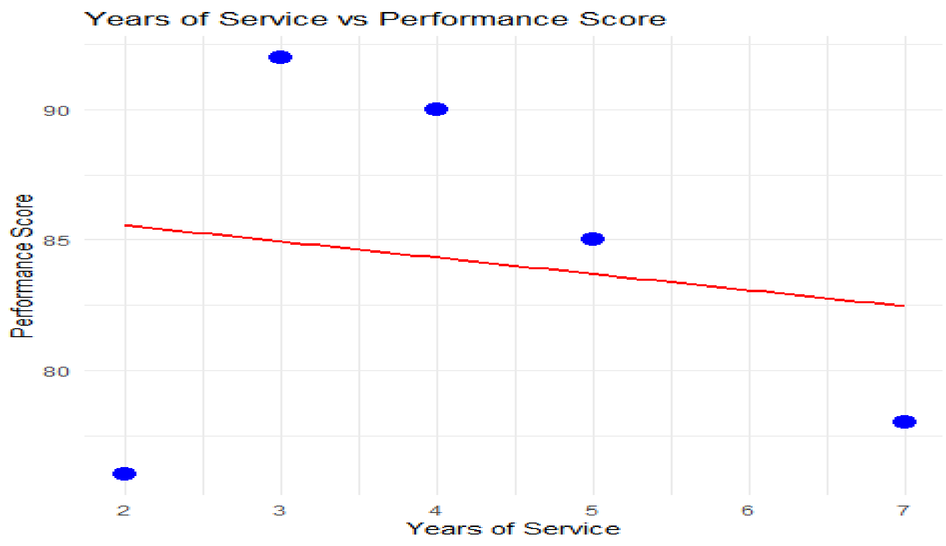
3C. Scatter Plot (Years of Service vs Performance Score)

Program:

```
library(ggplot2)
ggplot(employee_data, aes(x=YearsOfService, y=PerformanceScore)) +
  geom_point(size=4, color="blue") +
  geom_smooth(method="lm", se=FALSE, color="red") +
  labs(
    title="Years of Service vs Performance Score",
    x="Years of Service",
    y="Performance Score"
  ) +
```

```
  theme_minimal()
```

Output:



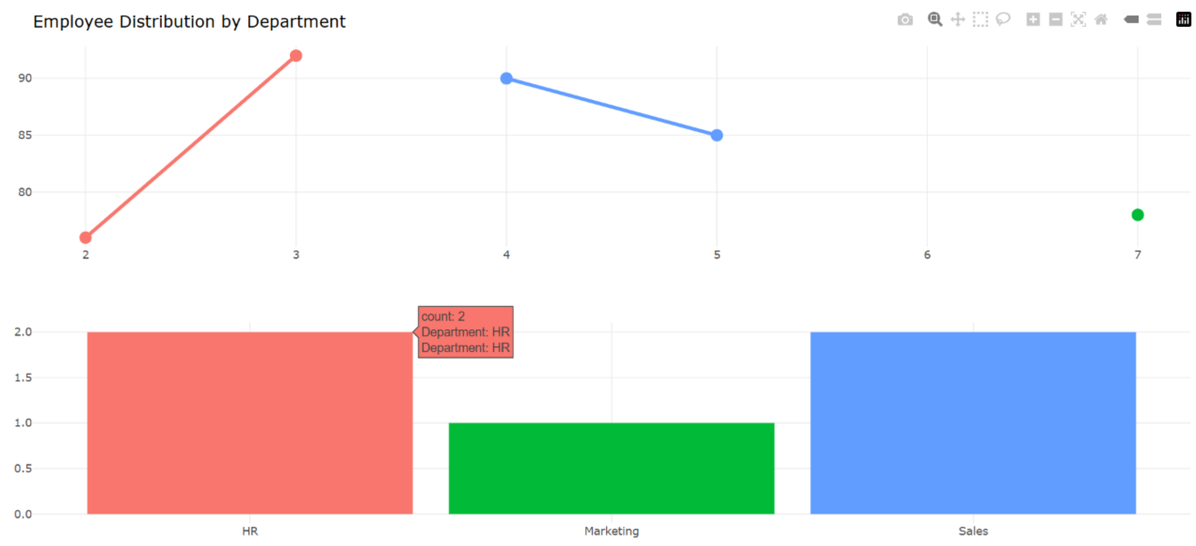
3D. Interactive Dashboard

Program:

```
library(plotly)
line_plot <- ggplotly(
  ggplot(employee_data,
         aes(x=YearsOfService,
             y=PerformanceScore,
             color=Department,
             group=Department)) +
    geom_line(size=1) +
    geom_point(size=3) +
    labs(title="Employee Performance Trend",
         x="Years of Service",
         y="Performance Score") +
    theme_minimal()
)
bar_plot <- ggplotly(
  ggplot(employee_data,
```

```
        aes(x=Department,
            fill=Department)) +
    geom_bar() +
    labs(title="Employee Distribution by Department",
        x="Department",
        y="Number of Employees") +
    theme_minimal() +
    theme(legend.position="none")
)
dashboard <- subplot(line_plot, bar_plot, nrows=2, margin=0.08)
dashboard
```

Output:



Set 4: Product Inventory Management

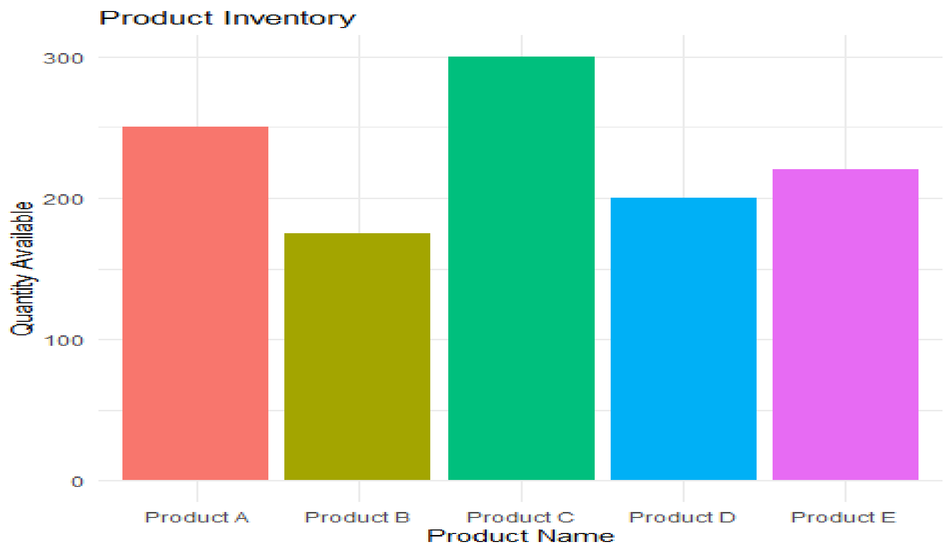
4A. Bar Chart

Program:

```
inventory_data <- data.frame(
ProductID=c(1,2,3,4,5),
ProductName=c("Product A","Product B","Product C","Product D","Product E"),
QuantityAvailable=c(250,175,300,200,220)
)
```

```
ggplot(inventory_data,
aes(x=ProductName,
y=QuantityAvailable,
fill=ProductName))+
geom_bar(stat="identity")+
labs(title="Product Inventory",
x="Product Name",
y="Quantity Available")+
theme_minimal()+
theme(legend.position="none")
```

Output:



4B. Stacked Bar Chart

Program:

```
inventory_data$Category <- c(
"Electronics",
"Electronics",
"Accessories",
"Accessories",
"Home"
)
ggplot(inventory_data,
```

aes(x=Category,

y=QuantityAvailable,

fill=ProductName))+

geom_bar(stat="identity")+

labs(title="Quantity by Product Category",

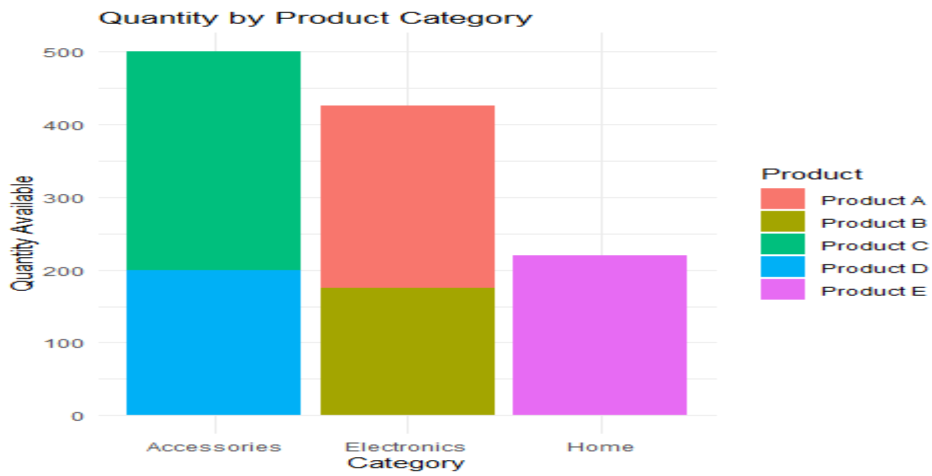
x="Category",

y="Quantity Available",

fill="Product")+

theme_minimal()

Output:



4C. Scatter Plot

Program:

inventory_data$Price <- c(500,350,650,450,550)

ggplot(inventory_data,

aes(x=Price,

y=QuantityAvailable))+

geom_point(size=4,color="blue")+

geom_smooth(method="lm",se=FALSE,color="red")+

labs(title="Price vs Quantity Available",

x="Product Price",

y="Quantity Available")+

theme_minimal()

Output:

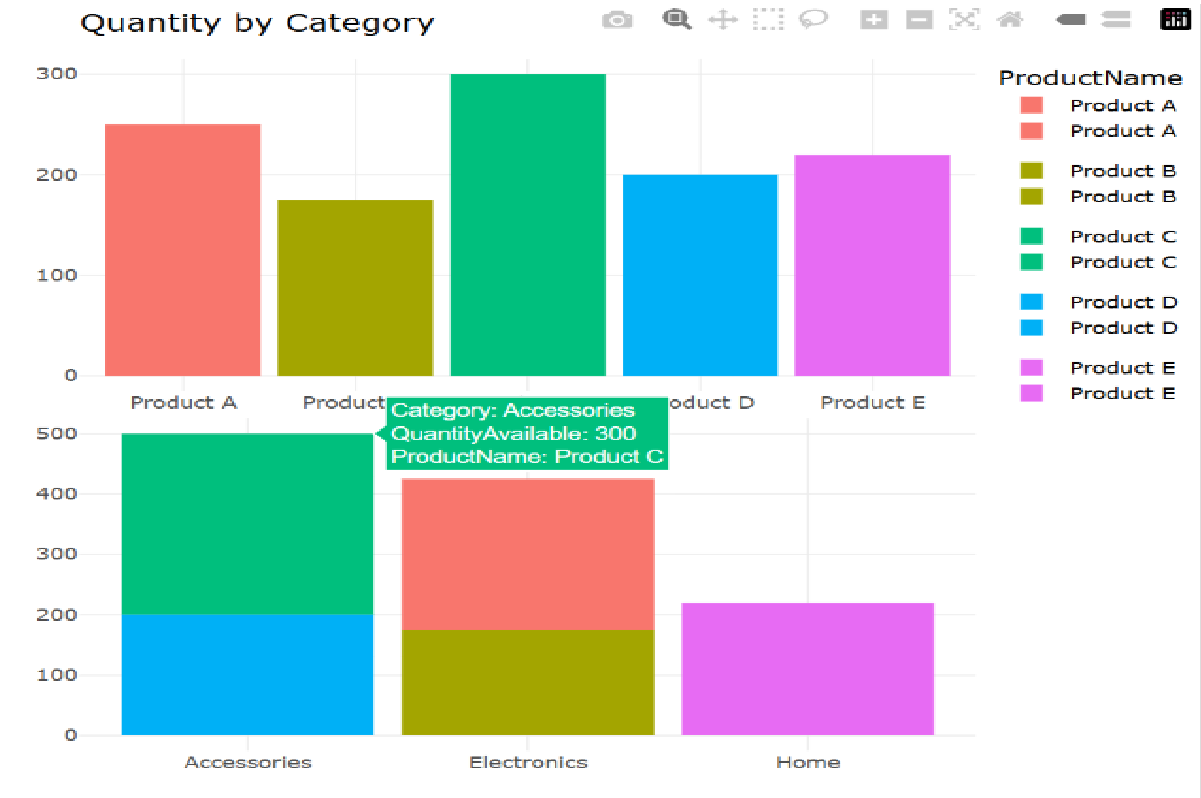
4D. Interactive Dashboard (R)

Program:

```
library(plotly)
bar_plot <- ggplotly(
ggplot(inventory_data,
aes(x=ProductName,
y=QuantityAvailable,
fill=ProductName))+
geom_bar(stat="identity")+
labs(title="Product Inventory")+
theme_minimal()+
theme(legend.position="none")
)
stacked_plot <- ggplotly(
ggplot(inventory_data,
aes(x=Category,
y=QuantityAvailable,
fill=ProductName))+
geom_bar(stat="identity")+
labs(title="Quantity by Category")+
```

```
theme_minimal()
)
dashboard <- subplot(bar_plot, stacked_plot, nrows=2)
dashboard
```

Output:



SET 5- Website Analytics

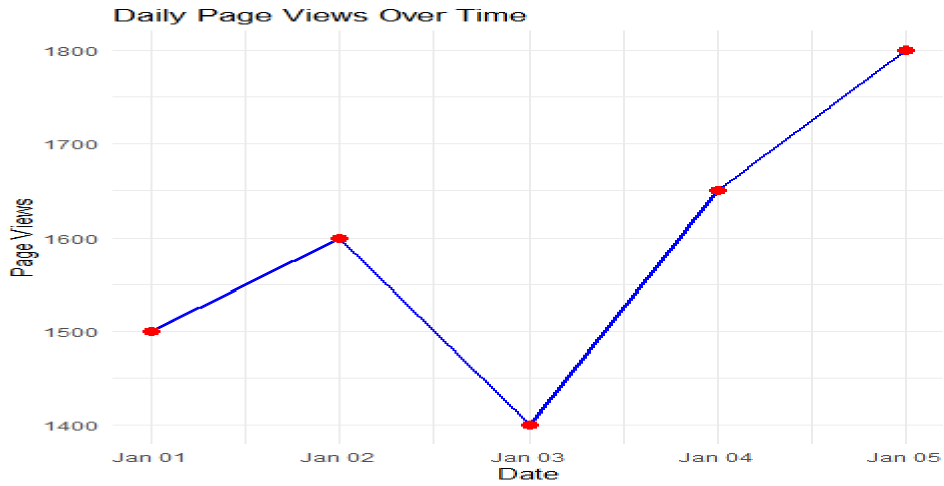
5A. Line Chart

Program:

```
website_data <- data.frame(
Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
PageViews=c(1500,1600,1400,1650,1800),
CTR=c(2.3,2.7,2.0,2.4,2.6)
)
ggplot(website_data,
aes(x=Date,y=PageViews,group=1))+
geom_line(color="blue",size=1)+
geom_point(color="red",size=3)+
```

```
labs(title="Daily Page Views Over Time",
x="Date",
y="Page Views")+
theme_minimal()
```

Output:

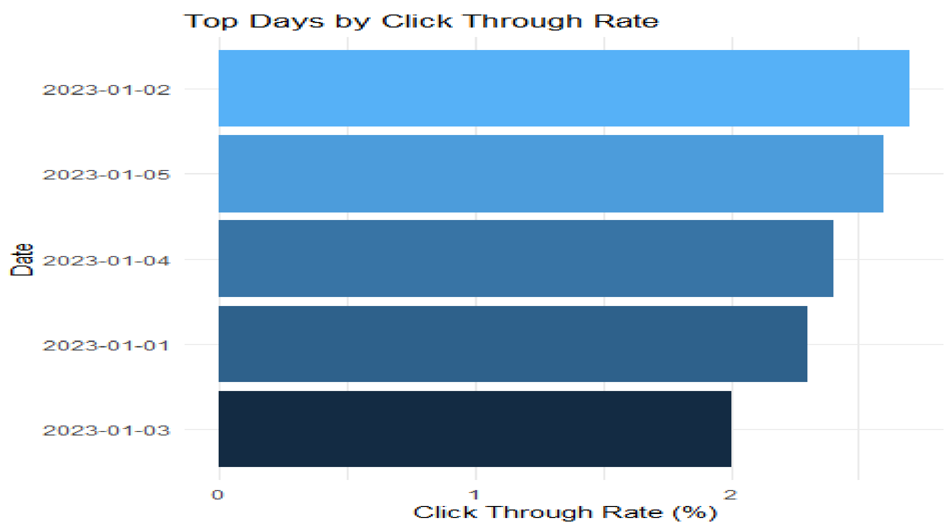


5B. Bar Chart (Top N Days with Highest CTR)

Program:

```
top_ctr <- website_data %>%
arrange(desc(CTR))
ggplot(top_ctr,
aes(x=reorder(as.character(Date),CTR),
y=CTR,
fill=CTR))+
geom_bar(stat="identity")+
coord_flip()+
labs(title="Top Days by Click Through Rate",
x="Date",
y="Click Through Rate (%)")+
theme_minimal()+
theme(legend.position="none")
```

Output:



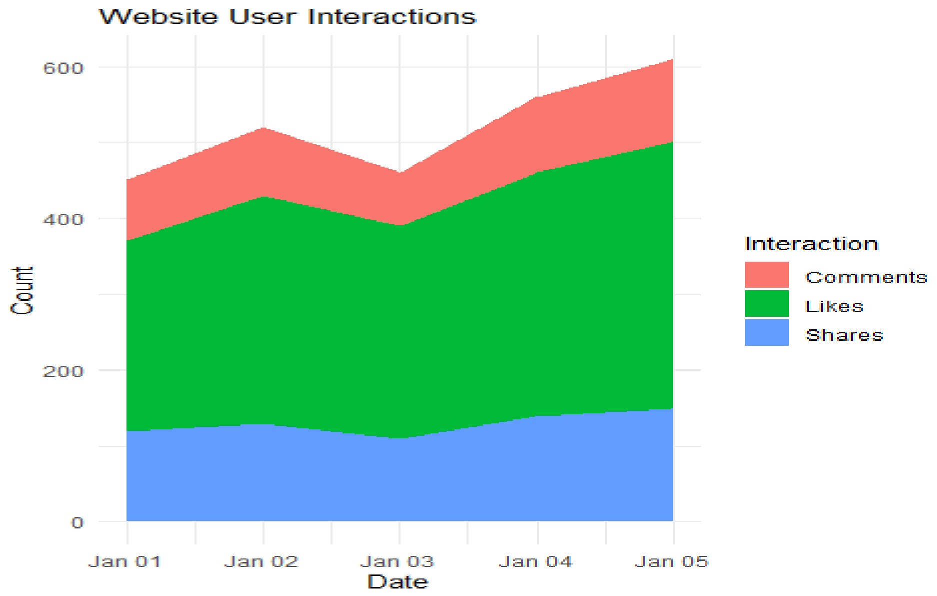
5C. Stacked Area Chart

Program:

```
website_data$Likes <- c(250,300,280,320,350)

website_data$Shares <- c(120,130,110,140,150)

website_data$Comments <- c(80,90,70,100,110)

interaction_data <- data.frame(

Date=rep(website_data$Date,3),

Interaction=c(rep("Likes",5),

rep("Shares",5),

rep("Comments",5)),

Count=c(website_data$Likes,

website_data$Shares,

website_data$Comments)

)

ggplot(interaction_data,

aes(x=Date,

y=Count,

fill=Interaction))+

geom_area()+

labs(title="Website User Interactions",
```

```
x="Date",
y="Count")+
theme_minimal()
```

Output:



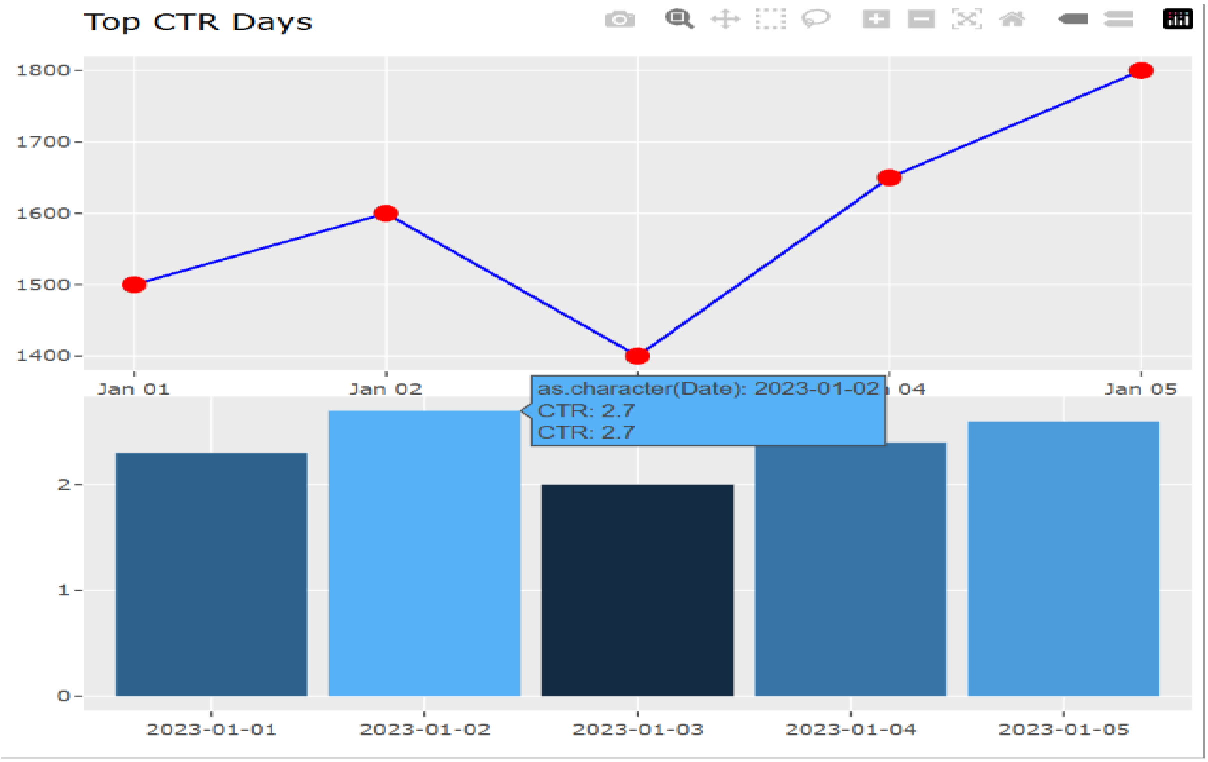
5D. Interactive Dashboard

Program:

```
line_plot <- ggplotly(
ggplot(website_data,
aes(x=Date,y=PageViews,group=1))+
geom_line(color="blue")+
geom_point(color="red",size=3)+
labs(title="Daily Page Views")
)
bar_plot <- ggplotly(
ggplot(top_ctr,
aes(x=as.character(Date),y=CTR,fill=CTR))+
geom_bar(stat="identity")+
labs(title="Top CTR Days")+
theme(legend.position="none")
)
```

```
dashboard <- subplot(line_plot,bar_plot,nrows=2)
dashboard
```

Output:



SET 6- Product Sales Analysis

6A. Grouped Bar Chart

Program:

```
sales_data <- data.frame(
ProductID=c(1,2,3),
ProductName=c("Product A","Product B","Product C"),
January=c(2000,1500,1200),
February=c(2200,1800,1400),
March=c(2400,1600,1100)
)
sales_long <- melt(sales_data,
id.vars="ProductName",
variable.name="Month",
value.name="Sales")
```

ggplot(sales_long,

aes(x=ProductName,

y=Sales,

fill=Month))+

geom_bar(stat="identity",position="dodge")+

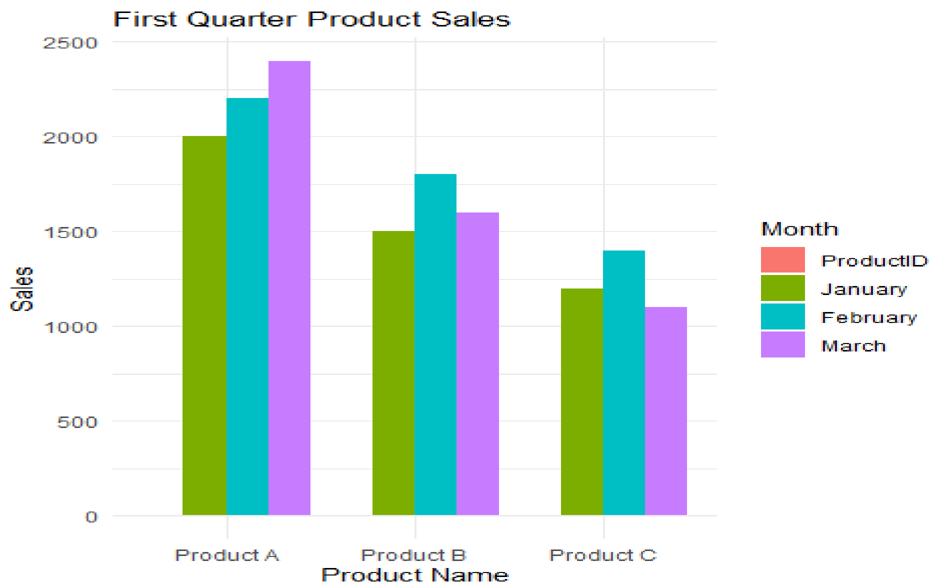
labs(title="First Quarter Product Sales",

x="Product Name",

y="Sales")+

theme_minimal()

Output:



6B. Stacked Area Chart

Program:

ggplot(sales_long,

aes(x=Month,

y=Sales,

fill=ProductName,

group=ProductName))+

geom_area()+

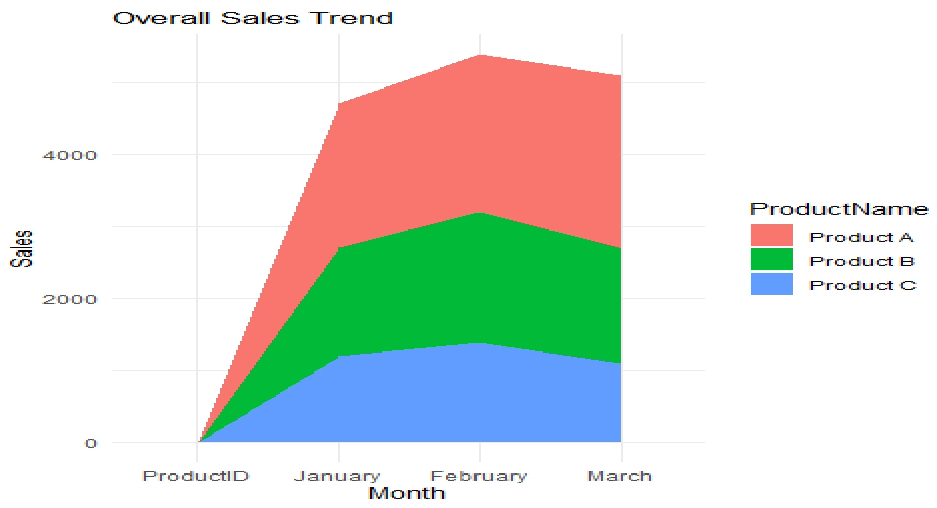
labs(title="Overall Sales Trend",

x="Month",

y="Sales")+

theme_minimal()

Output:



6C. Monthly Sales Table

Program:

sales_data

knitr::kable(

sales_data,

caption="Monthly Product Sales"

)

Output:

```
Table: Monthly Product Sales

| ProductID|ProductName | January| February| March|
|---------:|:-----------|-------:|--------:|-----:|
|         1|Product A   |    2000|     2200|  2400|
|         2|Product B   |    1500|     1800|  1600|
|         3|Product C   |    1200|     1400|  1100|
>
```

6D. Interactive Dashboard

Program:

bar_plot <- ggplotly(

ggplot(sales_long,

aes(x=ProductName,

```
y=Sales,
fill=Month))+
geom_bar(stat="identity",position="dodge")+
labs(title="Quarterly Product Sales")
)
area_plot <- ggplotly(
ggplot(sales_long,
aes(x=Month,
y=Sales,
fill=ProductName,
group=ProductName))+
geom_area()+
labs(title="Overall Sales Trend")
)
dashboard <- subplot(bar_plot, area_plot, nrows=2)
dashboard
```

Output: